

LEO AgriTourism Station - Mass Budget Calculations

Based on NASA-STD-3001 Standards

Tourism Module Mass Budget

Mission Parameters:

- 6 Tourists + 4 Crew = 10 Persons
- Mission Duration: 7 days
- Orbit: 400-450km

1. Life Support Systems

Atmospheric Control:

- O₂ Requirements: 10 persons x 0.84 kg/day = 8.4 kg/day
- CO₂ Scrubbing: 10 persons x 1.04 kg/day = 10.4 kg/day
- System Mass: 2,500kg (based on ISS ECLSS scaling)

Water Management:

- Water Consumption: 10 persons x 3.5kg/day = 35 kg/day
 - 7-day-Reserve: 245kg
 - Recycling System: 1,800 kg
- Total water Systems: 2,045kg

Food Systems:

- Food Storage: 10 persons x 1.83 kg/day x 7 days = 128kg
 - Preparation Equipment: 450kg
- Total Food Systems: 578kg

Waste Management:

- Body Waste System: 800kg
 - Trash Management: 350kg
- Total Waste Systems: 1,150kg

2. Structural Components

Pressure Vessel:

- Based on Unity Node scaling
- Unity: 11.9 tons for basic node
- Tourism Module (larger): 14,500kg

Interior Fittings:

- Sleeping Quarters: 6 pods x 200kg = 1,200kg
- Common Areas: 2,800 kg

Total Interior: 4,000kg

Windows & Observation:

- Earth Viewing Windows: 800kg
- Window protection: 300kg

Total Windows: 1,100kg

3. Safety & Emergency Systems

Emergency Life Support:

- Backup O2 Systems: 600kg
- Emergency Water: 400kg

Total Emergency: 1,000kg

Fire Suppression:

- Fire Detection & Suppression: 350kg

Medical Equipment:

- Basic Medical Kit: 150kg

Tourism Module Total Mass

System	Mass (kg)
Life Support	6,273
Pressure Vessel	14,500
Interior Fittings	4,000
Windows	1,000
Safety Systems	1,500
Subtotal	27,373
Growth Margin (15%)	4,106
TOTAL TOURISM MODULE	31,479

Agriculture Module Mass Budget

Based on Destiny Lab scaling + NASA Aeroponics data

1. Aeroponic Systems

Growing Systems:

- Aeroponic Chambers: $40 \text{ m}^2 \times 25 \text{ kg/m}^2 = 1,000\text{kg}$
 - Nutrient Delivery: 800kg
- Total Growing: 1,800kg**

LED Lighting Arrays:

- LED Systems: $40 \text{ m}^2 \times 50\text{kg/m}^2 = 2,000\text{kg}$
 - Power Control: 500kg
- Total Lighting: 2,500kg**

Environmental Control:

- Atmosphere Management: 1,200kg
 - Temperature Control: 800kg
 - Humidity Control: 600kg
- Total Environmental: 2,600kg**

2. Structural Components

Pressure Vessel:

- Based on Destiny Lab: 12,000kg

Research Equipment:

- Scientific Instruments: 1,500kg
 - Data Systems: 800kg
- Total Research: 2,300kg**

Agriculture Model Total Mass

System	Mass (kg)
Aeroponic Systems	6,900
Pressure Vessel	12,000
Research Equipment	2,300
SUBTOTAL	21,200
Growth Margin (15%)	3,180
TOTAL AGRICULTURE MODULE	24,380

Complete Station Mass Summary

Module	Mass (kg)
Tourism Module	31,479
Agriculture Module	24,380
Central Hub	15,000
Power Module	18,000
Propulsion/Station-keeping	8,000
Assembly Hardware	5,000
TOTAL STATION MASS	101,859

Launch Requirements

Falcon 9 Specifications:

- LEO Capacity: 22,800kg
- Cost per Launch: \$62 Mio

Launch Analysis:

- Total Mass: 101,859kg
- Required Launches: $101,859 \div 22,800 = 4.47 \rightarrow 5$ launches
- Total transport Cost: $5 \times \$62\text{M} = \310M

REFERENCES

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NOTES ON CALCULATIONS:

1. Life Support scaling based on NASA-STD-3001 per-person consumption rates
2. Structural masses derived from ISS heritage hardware with 15% growth margin
3. Launch costs updated to 2024 SpaceX published rates (\$67M per Falcon 9)
4. All mass estimates include NASA-standard 15% growth margin for development